09 · Does the "€100 -> Gold tier" perk retain people? — RDD (CausalPy)

July 7, 2026

1 09 · Does the "€100 -> Gold tier" perk retain people? — RDD (CausalPy)

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Customers who cross **€100** annual spend get **Gold** status and its perks. Gold members retain better — but of course they do, they're big spenders. Does the *perk itself* boost retention, or are we just seeing that big spenders would retain anyway? And where should we set the threshold?

1.0.1 The idea: a sharp rule creates a natural experiment

Regression discontinuity (RDD) exploits the fact that Gold status is assigned by a **sharp rule** on a **running variable** (annual spend): below €100 you're out, at €100-or-above you're in. Now compare a customer at €99.50 with one at €100.50. On *retention potential* they are essentially identical — nobody is meaningfully more loyal for having spent one euro more. The *only* thing that differs between them is that one got the perk and one didn't. So **any jump in retention exactly at €100** must be caused by the perk, not by the underlying "big spenders retain more" trend (which is smooth through the cutoff). RDD reads off the size of that jump.

1.0.2 What it can and can't tell you

- It estimates a **LATE at the cutoff** — the perk's effect *for customers near €100*. It says nothing about your €500 whales; don't extrapolate the jump to them.
- It's only valid if customers can't **manipulate** their spend to just clear €100 (that would make the people just above and just below systematically different). We test this with the **McCrary density** test (a suspicious pile-up of customers just above the cutoff).
- A headline RD number is only as good as its **robustness checks**: does the jump survive different **bandwidths** (how wide a window around €100 we use), **placebo cutoffs** (fake thresholds should show no jump), and **polynomial order** (linear vs curved fits)? We run all of them.

> **On real data.** RDD fits anywhere a **threshold rule** assigns a treatment: loyalty tiers, credit-score cutoffs for an offer, "spend €X for free shipping," scholarship/eligibility scores. You need the running variable, the cutoff, and the outcome — all of which live in your CRM.

7-step contract.

1.1 2 · Simulate a ground truth

Retention rises smoothly with annual spend (big spenders retain more anyway) — **plus** a true **+14pp jump** exactly at €100 from the Gold perk. The smooth part is the confounder; the jump is the causal effect. Spend is concentrated around the cutoff so both sides are well-populated.

```
TRUE perk effect = +14% retention at €100  ·  50% are Gold
```

	spend	treated	retention
0	84.325206	0	1.0
1	101.996816	1	0.0
2	93.392393	0	1.0
3	93.281224	0	0.0
4	102.504635	1	0.0

1.2 3 · Identify — a jump at the cutoff is causal (and its validity conditions)

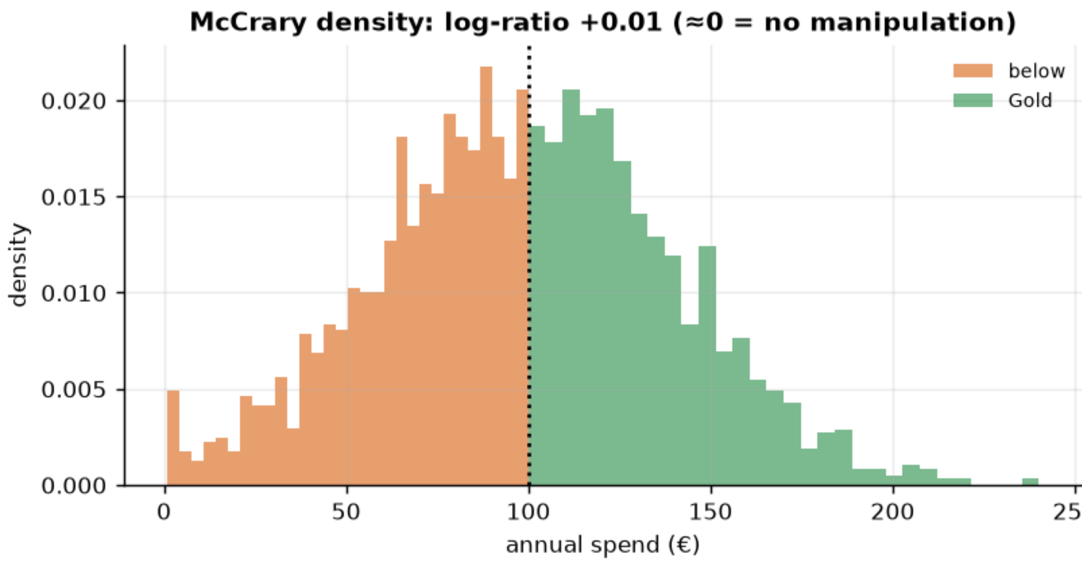
Sharp RD estimand: $\tau_{RD} = \lim_{x\downarrow c} \mathbb{E}[Y \mid X{=}x] - \lim_{x\uparrow c} \mathbb{E}[Y \mid X{=}x]$ — the discontinuity in retention at spend = €100. **Identification:** potential-outcome means are *continuous* at the cutoff, so units just either side are comparable and the jump is causal. This is a **LATE at the cutoff** — local to €100. Two things must hold, both checked below:

- **No manipulation** of the running variable at the cutoff (customers/business can't sort across it) — the McCrary density test.
- **The functional form isn't driving the jump** — checked via bandwidth, placebo cutoffs, and polynomial order.

1.3 3b · Validity check 1 — McCrary density (manipulation)

If customers game their spend to just clear €100 (or the business nudges them), the *density* of spend jumps at the cutoff and RD breaks. We compare the density just below vs just above.

```
density below 23.7 vs above 24.0, log-ratio +0.01 - no sign of sorting across
the cutoff.
```



(The density is smooth through €100 — no suspicious pile-up of customers *just* above the cutoff — so we have no evidence of manipulation, and the RDD comparison is fair.)

1.4 4 · Estimate — Bayesian regression discontinuity

Now the estimate itself. We fit a **local linear** regression on each side of the cutoff — a straight line for the below-€100 customers and another for the Gold customers — and the **height of the step where they meet at €100** is the causal effect of the perk. We restrict to a **bandwidth** of +/-€40 around the cutoff: the customers near the threshold are the comparable ones (a €98 and a €102 customer are alike; a €500 whale is not). Wider bandwidths borrow non-comparable customers; narrower ones throw away data — Step 5 checks the answer is stable to that choice.

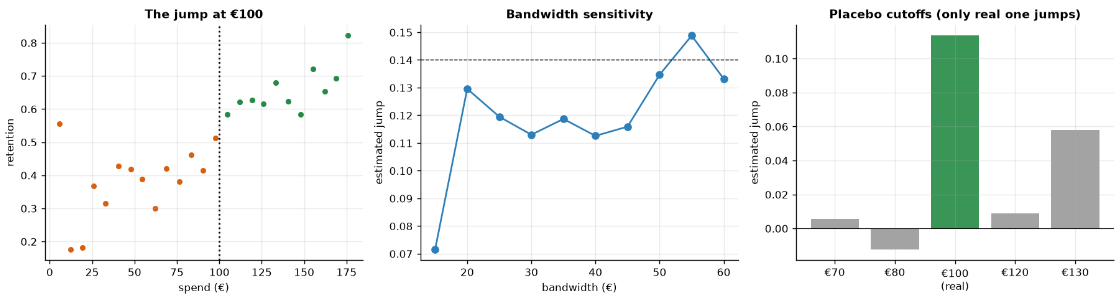
```
RD jump +11.3% (true +14%)  ·  90% CI [+2.8%, +19.2%]
```

1.5 5 · Validate — see the jump, plus bandwidth / placebo / polynomial robustness

A credible RD number is one that **survives** the researcher-degrees-of-freedom checks:

- **Bandwidth curve** — the estimate should be stable across a sensible range of bandwidths.
- **Placebo cutoffs** — run RD at fake thresholds (€70, €130) where there's no perk; the jump should be ~ 0. A real jump at a fake cutoff would mean the method finds discontinuities in noise.
- **Polynomial order** — linear vs quadratic local fit shouldn't move the answer much.

```
Bandwidth-stable around +12.0%.  Polynomial: linear +11.3%, quadratic +10.9%.
Placebo cutoffs: ['+0.6%', '+5.8%'] (~0).
```

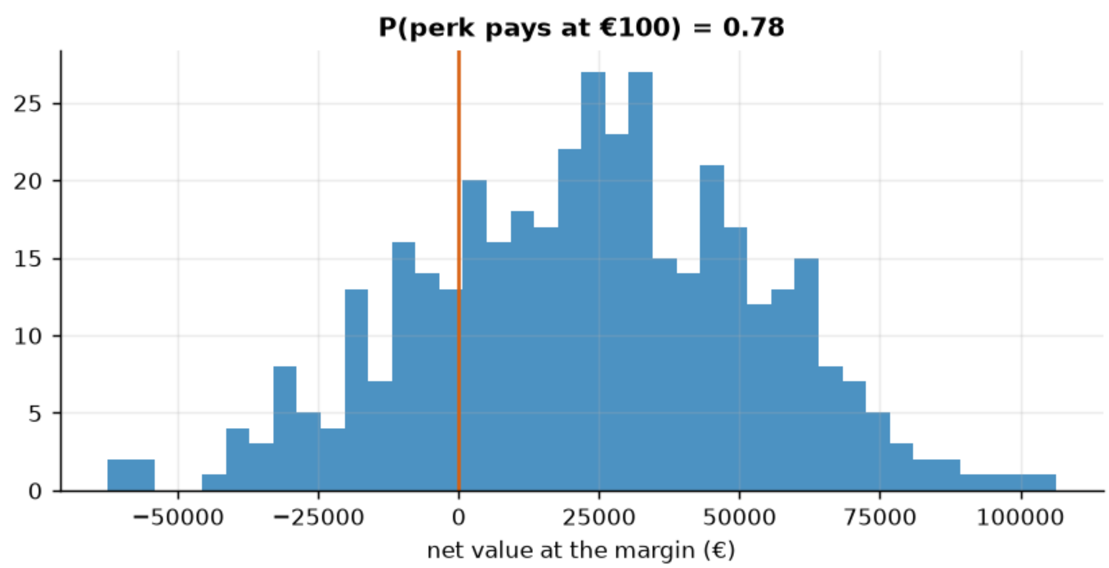


How to read the robustness battery. *Left* — bin retention by spend and you can *see* the step at €100 sitting on top of the smooth "big spenders retain more" trend; that step is the perk effect. *Middle* — the estimate barely moves as we vary the bandwidth, so we're not cherry-picking a window that happens to give a big number. *Right* — the decisive check: re-run the same RDD at **fake** cutoffs (€70, €80, €120, €130) where no perk exists; only the **real** €100 cutoff shows a jump, the placebos are ~ 0. A method that "finds" a discontinuity at €130 too would be finding noise — this one doesn't. Together with the McCrary density and the linear-vs-quadratic agreement, that's the full case that the jump is a real causal effect, not an artefact of how we drew the lines.

1.6 6 · Decide, in euros — does the perk pay, and where to set the cutoff?

Value the retention lift: extra retained customers x annual value, minus the perk cost, **at the current €100 margin** (that's what RD identifies). We also sweep the perk cost to find the break-even.

```
Net €7.7/member  ·  total €23,071 [90% €-28,191, €70,074]
P(perk pays) 0.78 -> reconsider  ·  break-even perk cost ~ €6/member
(conservative).
```



1.7 7 · Caveats

- **RD is *local*.** The effect is identified only *at the cutoff* — the perk's effect on customers near €100, not on your €500 whales. Don't extrapolate the jump to everyone.
- **Manipulation breaks it.** We checked the McCrary density; watch for heaping at round numbers.
- **Bandwidth is bias–variance.** We showed the estimate is stable across bandwidths; a fragile estimate that swings with bandwidth is a red flag.
- **Fuzzy RD if the rule isn't sharp.** If crossing €100 only *raises the probability* of Gold (grandfathering, overrides), use the fuzzy-RD ratio (jump in Y ÷ jump in P(treated)).